

WEBSTER NOLF, MD, USA

WMO: 725514

Lat: 38.142N

Lon: 76.429W

Elev: 21

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03757

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.6	20.6	-2.1	4.9	19.6	3.2	6.5	24.3	23.8	33.2	20.8	33.3	6.8	300	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	92.9	77.2	90.3	76.0	88.0	74.9	79.6	88.9	78.3	86.8	77.2	85.0	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	140.9	84.7	75.9	135.3	83.5	74.8	130.3	82.3	43.0	89.0	41.7	86.8	40.5	85.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.0	16.7	14.4	10.1	95.8	5.9	3.3	5.8	98.2	2.3	100.1	-1.0	101.9	-5.3	104.3		
			8.5	81.2	5.5	1.6	4.5	82.4	1.3	83.3	-1.8	84.2	-5.8	85.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	36.7	39.5	46.5	57.3	67.5	75.4	79.2	76.8	71.4	61.2	49.7	42.3	
	DBStd	16.75	9.92	9.92	9.74	8.50	7.55	5.52	4.79	4.69	6.20	8.26	7.88	8.70	
	HDD50	1343	429	319	189	27	0	0	0	0	0	8	100	271	
	HDD65	3847	877	715	577	258	65	2	0	0	14	173	463	703	
	CDD50	4529	16	24	79	246	543	763	906	832	643	354	89	34	
	CDD65	1559	0	0	2	27	143	315	441	367	206	54	3	1	
	CDH74	13590	0	0	22	228	983	2720	4541	3303	1498	291	4	0	
	CDH80	4519	0	0	1	52	268	917	1761	1041	420	59	0	0	
Wind	WSAvg	6.1	6.8	6.9	7.2	7.2	6.3	5.8	5.4	4.7	5.3	6.0	6.1	6.0	
Precipitation	PrecAvg	46.0	3.2	2.8	3.9	3.5	3.6	4.4	4.5	5.0	4.2	3.9	3.0	3.7	
	PrecMax	65.6	7.2	7.4	10.2	7.8	8.8	10.4	10.3	10.2	10.8	8.0	7.8	8.0	
	PrecMin	35.4	1.5	0.5	0.0	1.9	1.1	1.6	1.5	1.5	0.9	0.7	0.4	1.5	
	PrecStd	6.9	1.3	1.5	1.9	1.3	2.0	2.1	2.1	2.0	2.6	2.2	1.9	1.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	69.5	77.1	86.5	90.1	94.6	97.8	93.4	91.5	85.4	72.2	69.6	
		MCWB	56.9	59.3	61.9	67.3	74.8	76.2	78.7	78.3	75.3	72.7	63.9	63.9	
	2%	DB	61.3	64.4	71.1	80.3	86.2	91.1	93.8	90.4	88.0	80.8	68.9	65.3	
		MCWB	55.2	55.6	58.6	64.2	71.7	74.7	77.9	77.1	74.2	70.2	61.1	60.0	
	5%	DB	57.0	59.4	66.7	75.7	82.9	88.1	91.0	88.0	84.8	77.1	65.1	61.1	
		MCWB	52.0	52.8	56.3	63.1	70.6	74.2	77.2	75.5	72.4	68.8	58.9	57.1	
	10%	DB	52.5	54.6	61.8	71.3	79.6	85.4	88.2	85.6	81.7	73.8	62.1	56.0	
		MCWB	47.0	47.7	53.5	60.7	68.3	72.7	75.8	74.8	71.3	65.8	56.5	51.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.6	62.8	64.2	71.1	76.3	79.1	81.8	80.9	78.4	75.6	66.6	65.9
			MCDB	62.6	68.0	73.0	76.5	86.6	90.1	91.7	90.3	86.5	81.0	70.8	68.6
2%		WB	56.2	57.4	61.2	67.1	73.9	77.3	80.1	79.1	76.7	72.7	63.3	60.7	
		MCDB	60.4	61.8	69.2	75.6	82.7	86.7	89.6	87.6	84.0	78.3	66.6	64.0	
5%		WB	52.6	53.2	58.1	64.6	72.2	76.2	78.7	77.9	75.3	70.2	60.1	57.1	
		MCDB	56.9	58.8	65.0	73.8	80.0	84.8	87.6	85.3	81.1	75.8	64.0	60.4	
10%		WB	47.5	48.3	54.5	62.3	70.5	75.0	77.6	76.5	73.6	67.8	57.2	52.6	
		MCDB	51.9	54.0	60.4	69.9	77.6	82.8	85.7	83.2	79.4	72.8	61.5	55.4	
Mean Daily Temperature Range	5% DB	MDBR	14.6	16.1	17.2	18.6	16.8	16.4	15.6	15.5	15.9	16.8	16.1	15.0	
		MCDBR	19.7	21.5	23.0	23.1	19.1	18.7	17.6	16.4	18.3	17.8	18.7	17.9	
		MCWBR	17.1	16.0	13.9	11.1	9.1	7.7	6.8	7.0	8.4	9.4	12.8	15.8	
	5% WB	MCDBR	18.4	20.1	21.1	19.8	16.5	16.8	15.5	14.4	14.2	15.6	15.5	17.2	
		MCWBR	17.2	16.1	14.5	11.4	8.5	7.6	7.0	7.0	7.7	9.3	12.4	16.2	
Clear-Sky Solar Irradiance	taub		0.310	0.319	0.353	0.393	0.434	0.484	0.516	0.508	0.410	0.359	0.326	0.312	
	taud		2.518	2.485	2.432	2.336	2.257	2.162	2.088	2.095	2.347	2.490	2.542	2.550	
	Ebn at Noon		278	289	288	281	270	256	246	245	266	271	270	269	
	Edh at Noon		25	29	34	39	43	48	51	49	36	29	24	23	
All-Sky Solar Radiation	RadAvg		704	961	1283	1620	1843	2020	1953	1752	1417	1085	797	621	
	RadStd		54	102	107	115	199	148	94	95	140	120	69	53	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (79 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+155	

Nomenclature: See separate page